Pod (2)

1) NameSpace

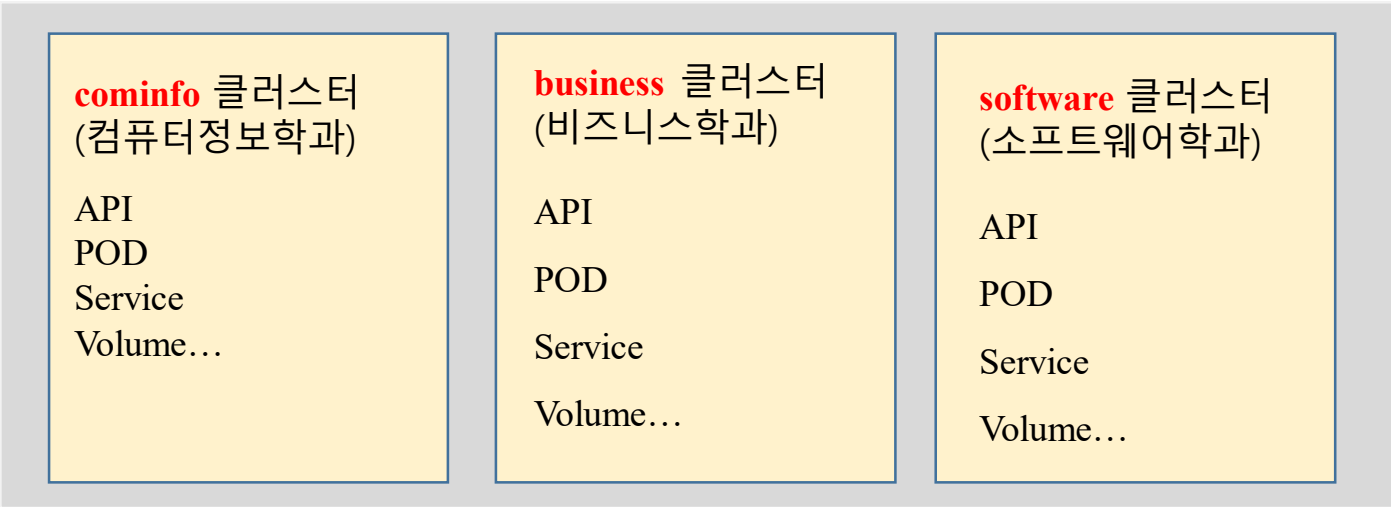
2) Init Container

3) Static Pod

1) NameSpace

NameSpace

- K8S API 종류 중 하나
- K8S cluster 하나를 여러 개 논리적인 단위로 나눠 사용하는 것
- K8S Cluster 하나를 여러 개 팀이나 사용자가 함께 공유 할 수 있음

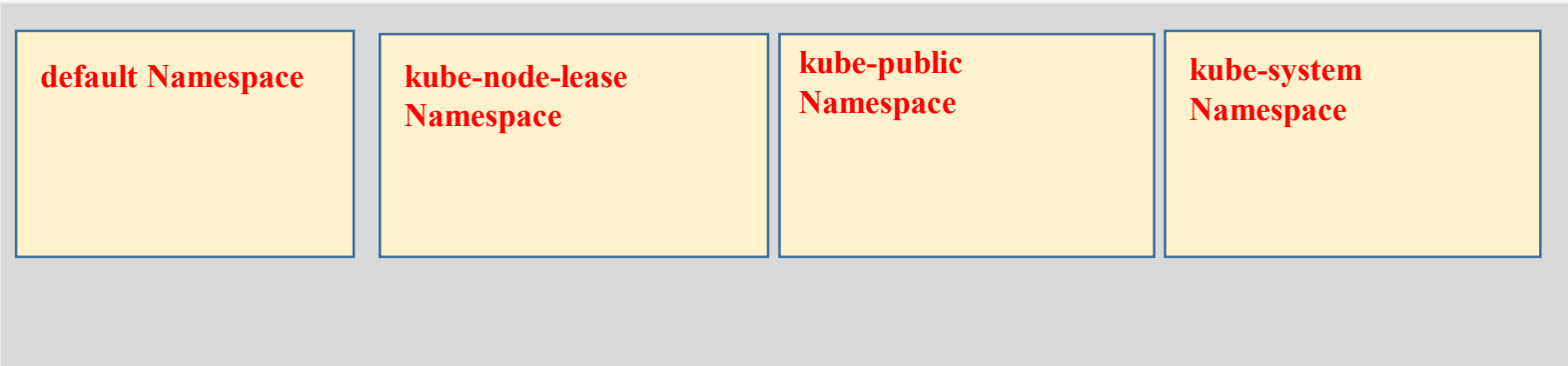


물리적 K8S 클러스터

현재 생성된 NameSpace 확인

kubectl get namespaces

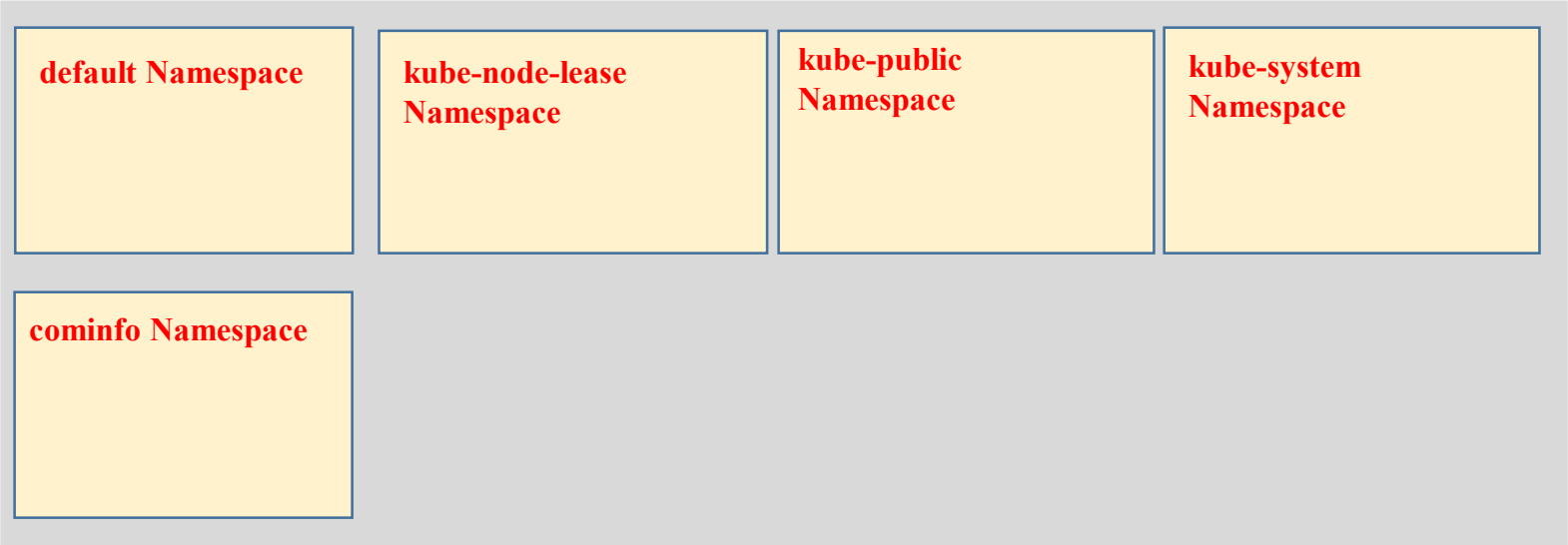
```
Terminal Host 1
controlplane $ kubectl get namespaces
NAME              STATUS   AGE
default           Active   42s
kube-node-lease   Active   43s
kube-public       Active   43s
kube-system       Active   43s
controlplane $
```



❶ NameSpace 생성

kubectl create namespace **cominfo**

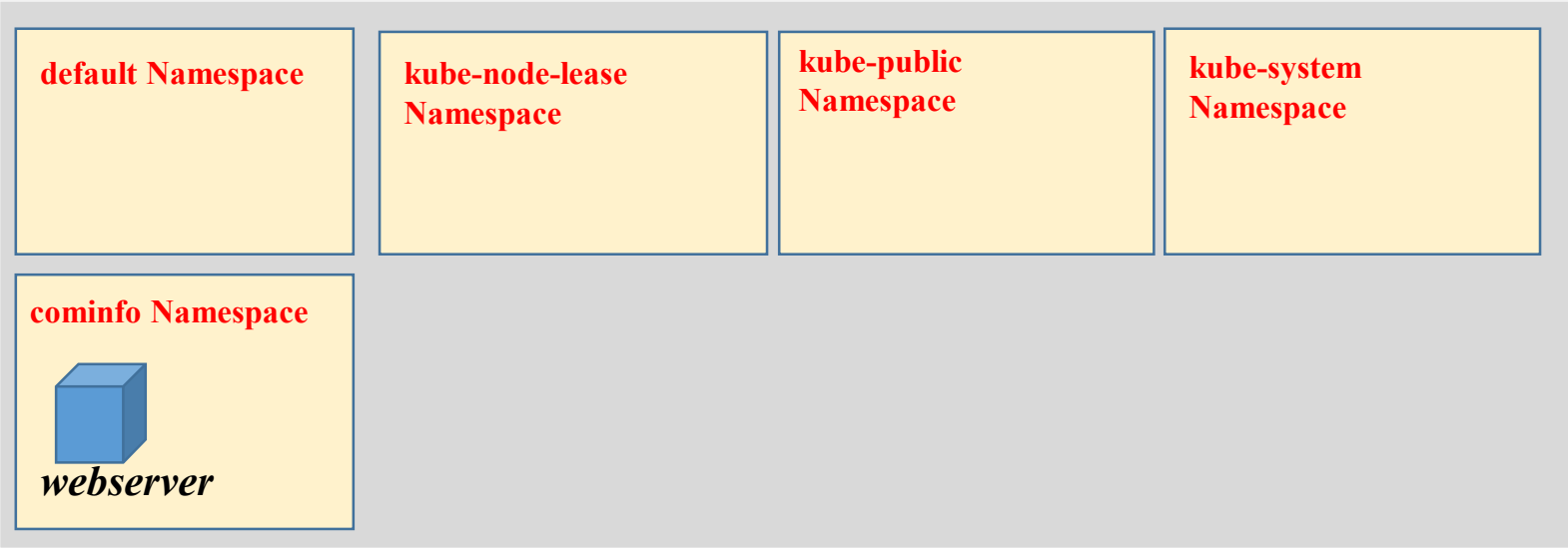
```
controlplane $ kubectl create namespace cominfo
namespace/cominfo created
controlplane $ kubectl get namespaces
NAME              STATUS   AGE
cominfo           Active   11s
default           Active   28m
kube-node-lease   Active   28m
kube-public       Active   28m
kube-system       Active   28m
controlplane $
```



❷ 지정된 NameSpace에 pod 생성

kubectl run webserver –image=nginx:1.14 –port 80 --namespace **cominfo**
kubectl get pods --namespace cominfo

```
controlplane $ kubectl run webserver --image=nginx:1.14 --port 80 --namespace cominfo
pod/webserver created
controlplane $
controlplane $ kubectl get pods --namespace cominfo
NAME        READY   STATUS    RESTARTS   AGE
webserver   1/1     Running   0          30s
controlplane $
```



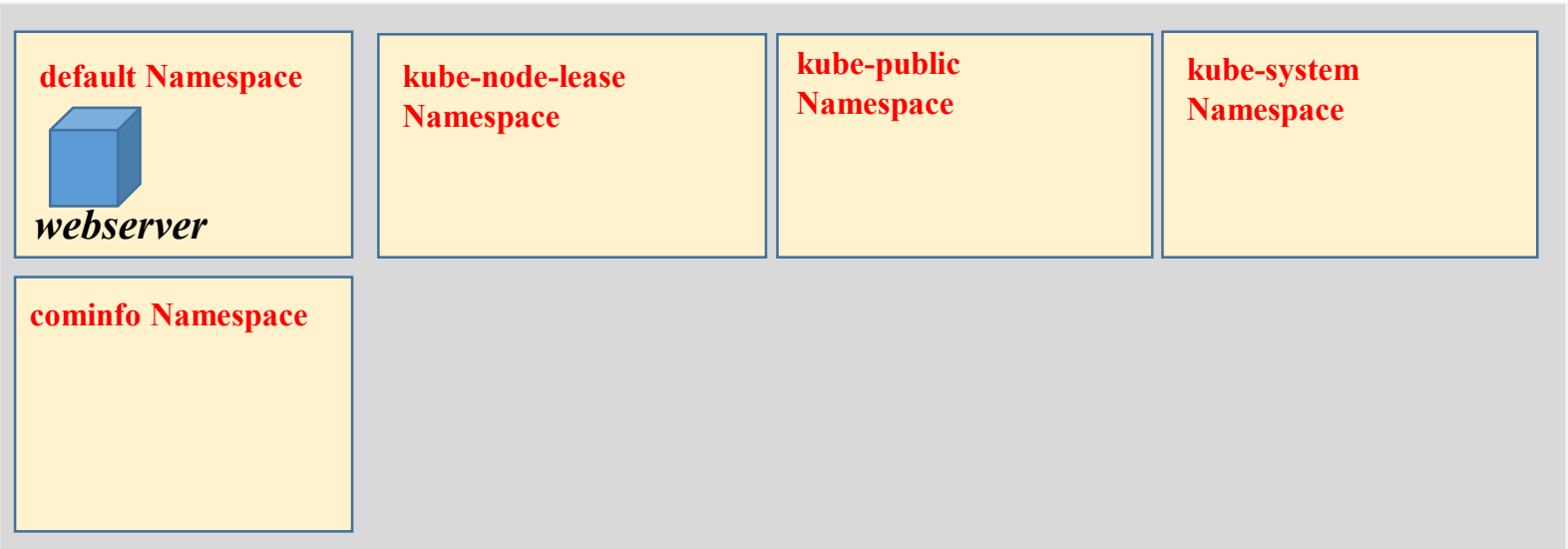
기본 namespace

kubectl run web --image=nginx:1.14 --port 80

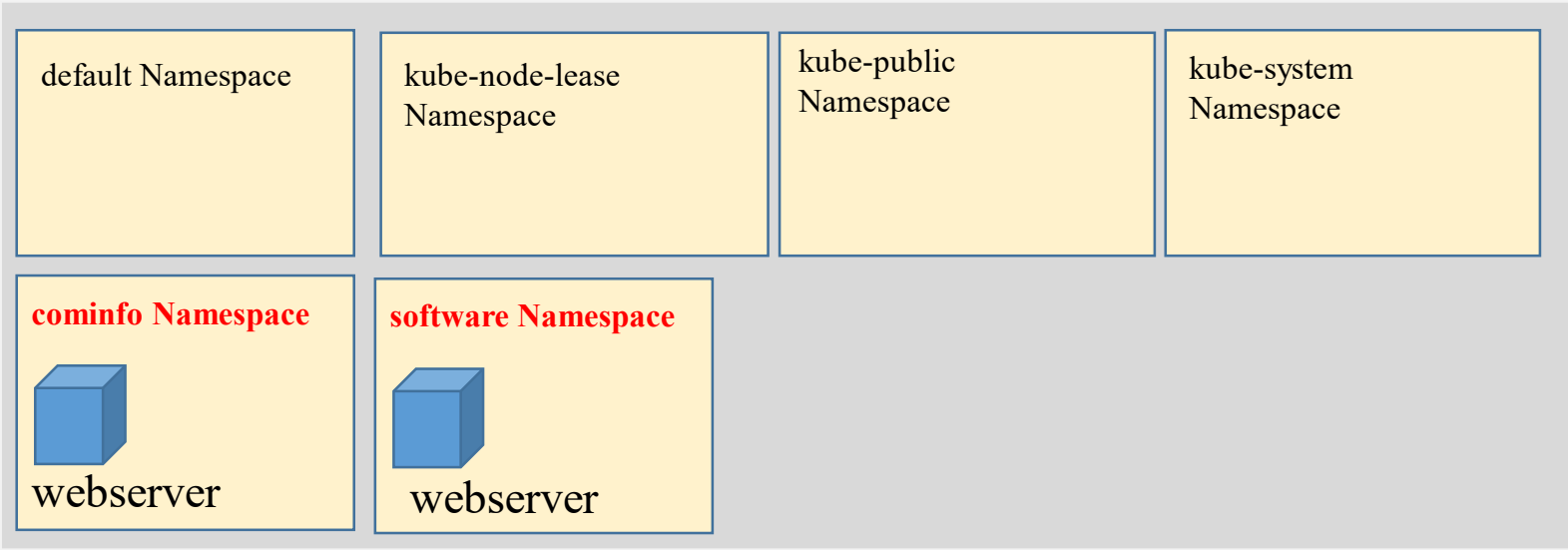
```
controlplane $ kubectl run web --image=nginx:1.14 --port 80
pod/web created
controlplane $ kubectl get pods
NAME    READY    STATUS     RESTARTS    AGE
web     1/1      Running    0           10s
```

kubectl get pod

```
controlplane $ kubectl get pod
No resources found in default namespace.
controlplane $
```



실습 1. Namespace 생성 후 지정된 cluster에 nginx server 생성 배치



- software Namespace 생성
- 생성된 software namespace에 nginx:1.14 Webserver pod 배치

2) Init Container

초기화 컨테이너(init container)

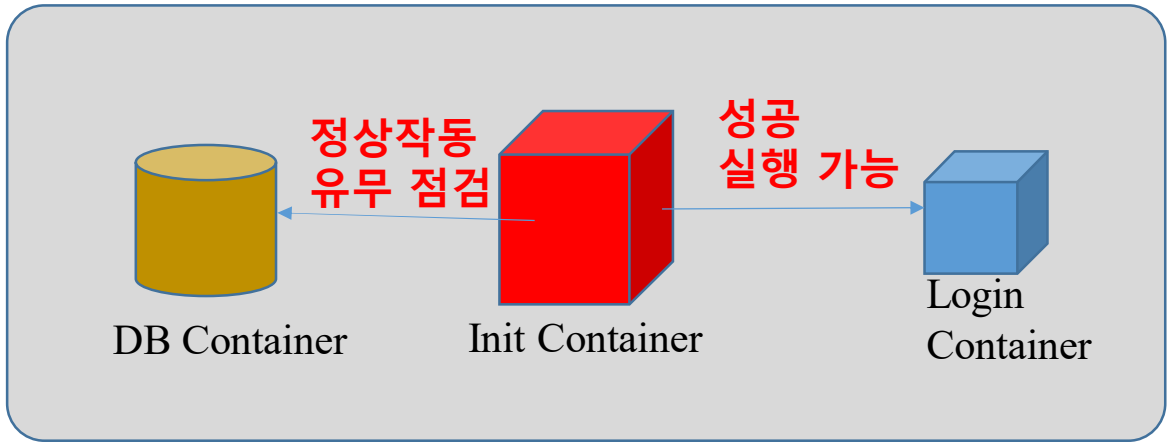
- Main container(App container)가 실행되기 전 동작시킨 컨테이너
- Main Container가 실행되기 전에 사전 작업이 필요한 경우 사용
- 초기화 컨터이너가 모두 실행된 후에 App container를 실행

  - Init container는 여러 개 구성할 수 있음

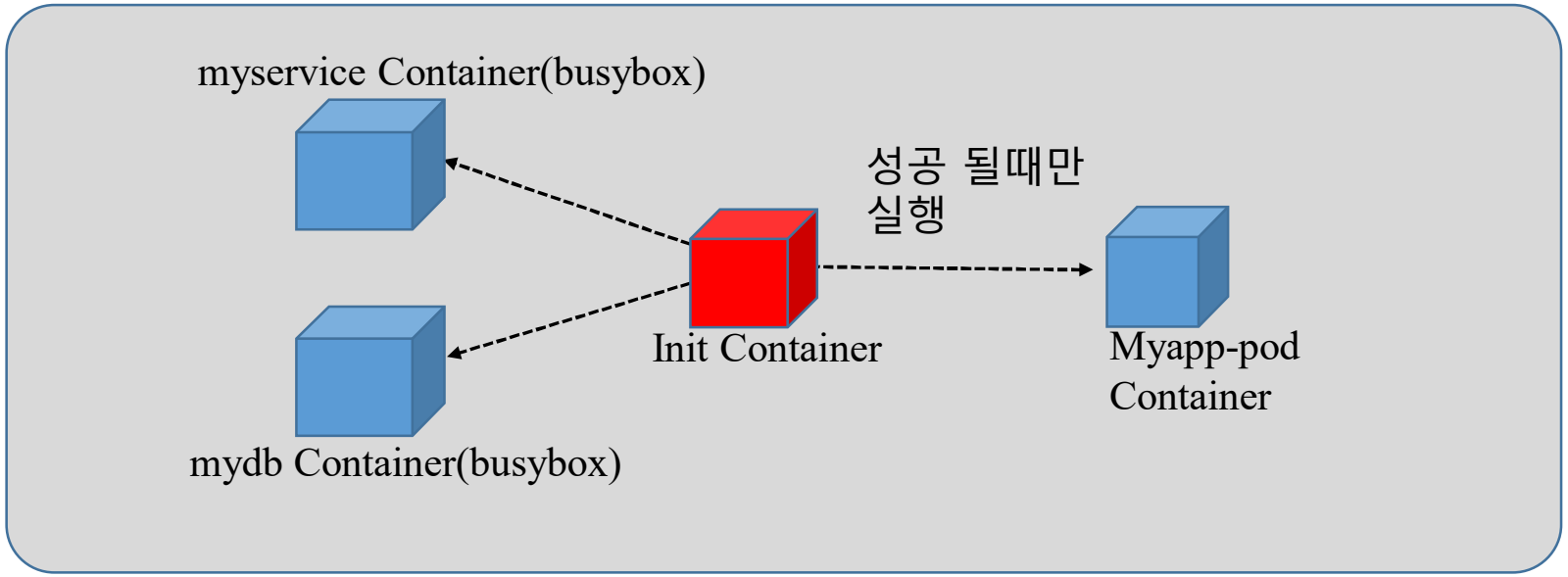
  - Init container 실행이 실패하면 성공할 때까지 재시작함

  - Init container가 모두 실행 된 후 App Container 실행이 시작

초기화 컨테이너(init container)

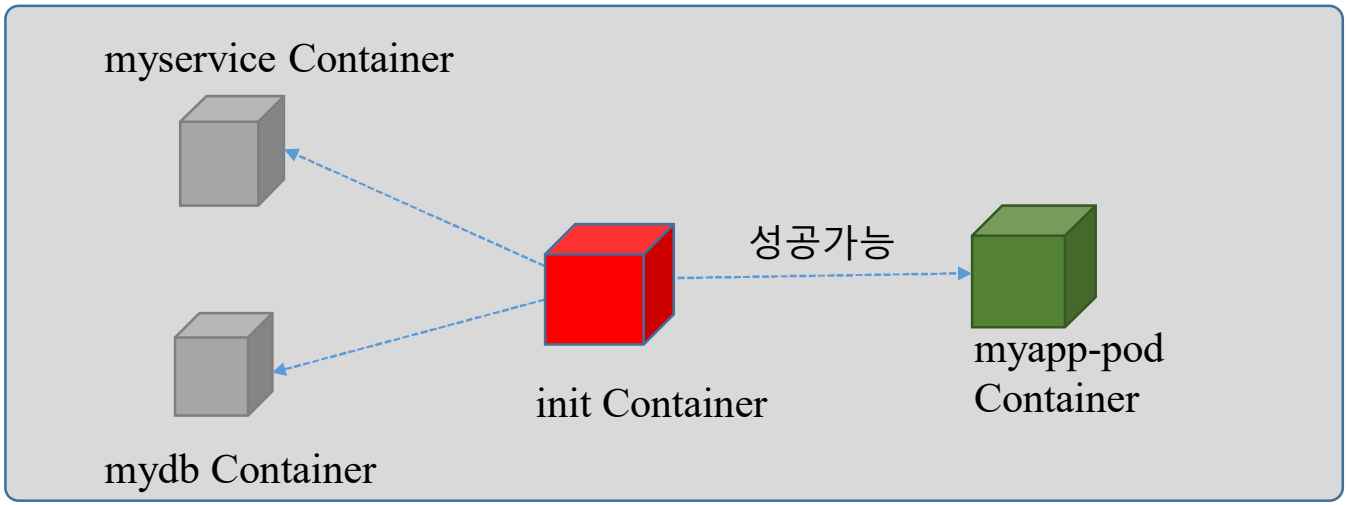


Pod



Alt+Enter

```
[node1 ~]$ kubectl get pods -o wide
NAME        READY   STATUS     RESTARTS   AGE   IP         NODE    NOMINATED NODE   READINESS GATES
myapp-pod   0/1     Init:0/2   0          48s   10.5.1.5   node2   <none>           <none>
[node1 ~]$
```




```
apiVersion: v1
kind: Service
metadata:
  name: myservice
spec:
  ports:
  - protocol: TCP
    port: 80
    targetPort: 9376
```

myservice.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: mydb
spec:
  ports:
  - protocol: TCP
    port: 80
    targetPort: 9377
```

mydb.yaml

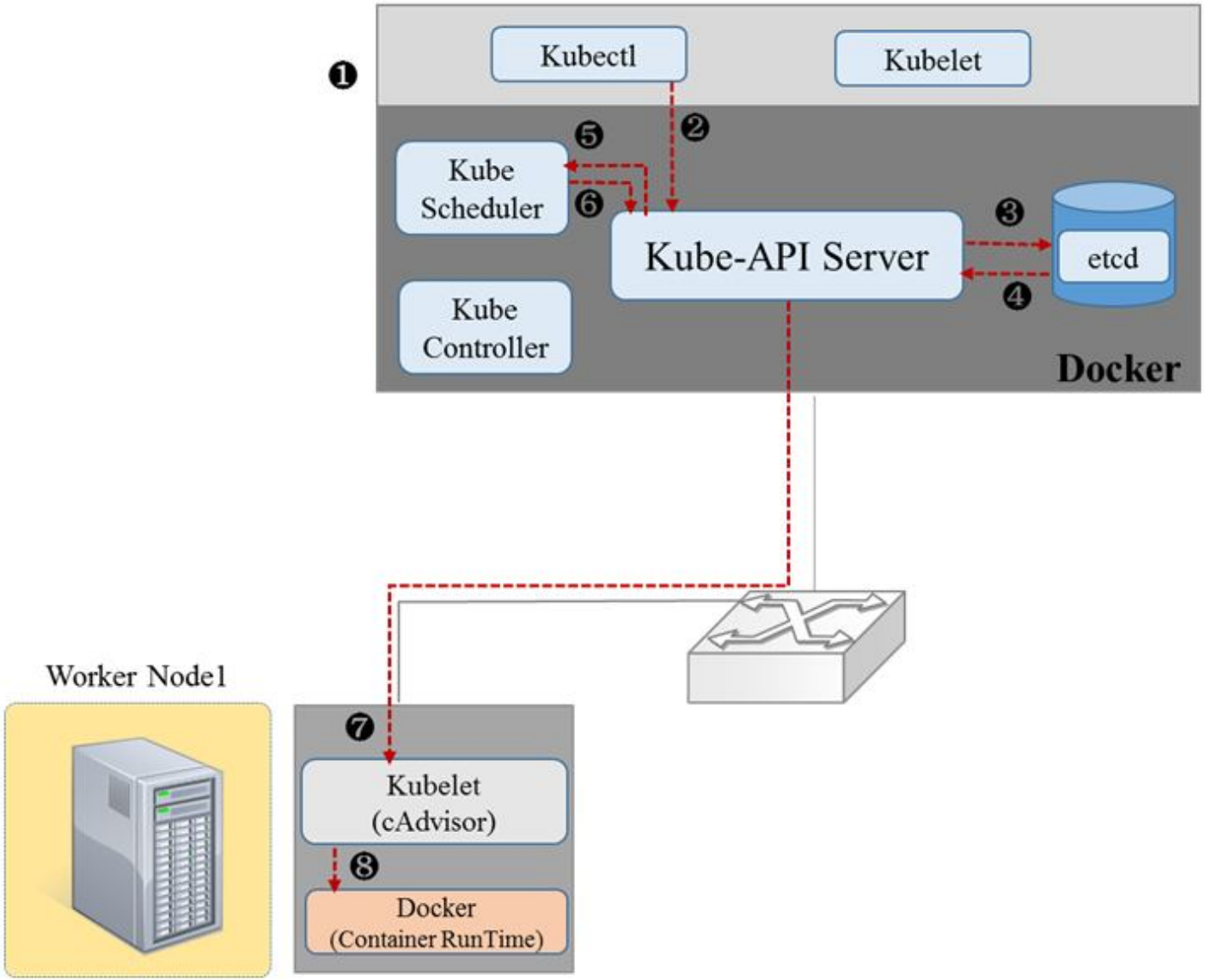
myapp-pod.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
spec:
  containers:
   - name: myapp-container
     중간 생략~~~
  initContainers:
   - name: myservice
     중간 생략~~~
   - name: mydb
     중간 생략~~~
```

init Container

- kubectl delete pod --all
- kubectl get pods –o wide --watch
- kubectl create –f init-container-exam.yaml
- kubectl create –f init-container-exam-svc.yaml
- kubectl create –f init-container-exam-db.yaml
- kubectl get pods - o wide
- kubectl describe pods *pod 명*

3) Static Pod


❶
Kubectl
Kubelet
❷
❺
Kube
Scheduler
❻
Kube-API Server
❸
etcd
❹
Kube
Controller
Docker
Worker Node1
❼
Kubelet
(cAdvisor)
❽
Docker
(Container RunTime)

Static Pod

- Kube-apiserver를 통하지 않고 kubelet이 직접 실행하는 pod
  - API 서버 없이 특정 노드에 있는 kubelet 데몬에 의해 직접관리
  - kubelet daemon에 의해 관리되는 container
- Kubelet이 직접 관리하면서 이상이 생기면 재시작

실습. Static Pod

```
[node1 ~]$ cd /var/lib/kubelet
[node1 kubelet]$ ls
config.yaml  cpu_manager_state  device-plugins  kubeadm-flags.env
[node1 kubelet]$ cat config.yaml
apiVersion: kubelet.config.k8s.io/v1beta1
authentication:
```

```
kind: KubeletConfiguration
logging: {}
nodeStatusReportFrequency: 0s
nodeStatusUpdateFrequency: 0s
rotateCertificates: true
runtimeRequestTimeout: 0s
shutdownGracePeriod: 0s
shutdownGracePeriodCriticalPods: 0s
staticPodPath: /etc/kubernetes/manifests
streamingConnectionIdleTimeout: 0s
syncFrequency: 0s
volumeStatsAggPeriod: 0s
[node1 kubelet]$
```

Pod lifecycle

- Pod 생성부터 삭제까지의 과정에서 생명주기(lifecycle)가 있음

kubectl get pods –o wide --watch

```
[node1 ~]$ kubectl get pods -o wide
NAME        READY   STATUS    RESTARTS   AGE   IP          NODE    NOMINATED NODE   READINESS GATES
multiple    2/2     Running   0          19s   10.5.1.3    node2   <none>           <none>
single      1/1     Running   0          11m   10.5.2.2    node3   <none>           <none>
```

$kubectl describe pods *nginx-pod-live*

```
[node1 ~]$ kubectl describe pods nginx-pod-live
Name:         nginx-pod-live
Namespace:    default
Priority:     0
Node:         node3/192.168.0.11
Start Time:   Sun, 01 May 2022 04:19:01 +0000
Labels:       <none>
Annotations:  <none>
Status:       Running
IP:           10.5.2.3
IPs:
  IP:  10.5.2.3
Containers:
```

Pod 생명주기(lifecycle)

Pending	K8S 시스템에 pod를 생성하는 중임을 뜻함 Container 이미지를 다운로드한 후 전체 container를 실행하는 도중 Pod안에 전체 container가 실행될때까지 시간이 걸림
Running	Pod 안에 모든 container가 실행 중인 상태 1개 이상의 container가 실행 중이거나 시작 또는 재시작 상태 임
Succeeded	Pod 안 모든 container가 정상 실행 종료된 상태로 재시작 되지 않았음
Failed	Pod 안 모든 container가 정상적으로 실행 종료되지 않은 container가 있는 상태 Container 종료코드가 0이 아니면 비정상종료 또는시스템이 직접 Container를 종료한것
Unknown	Pod의 상태를 알 수 없는 상태 (Pod가 있는 노드와 통신 할 수 없을 때)

```
[node1 kubelet]$ kubectl get pods
NAME        READY   STATUS     RESTARTS   AGE
myapp-pod   0/1     Init:0/2   0          11s
[node1 kubelet]$ kubectl describe pods myapp-pod
Name:         myapp-pod
Namespace:    default
Priority:     0
Node:         node2/192.168.0.12
Start Time:   Tue, 03 May 2022 04:01:37
Labels:       app=myapp
Annotations:  <none>
Status:       Pending
IP:           10.5.1.2
IPs:
  IP:  10.5.1.2
Init Containers:
```

```
Conditions:
  Type              Status
  Initialized       False
  Ready             False
  ContainersReady   False
  PodScheduled      True
Volumes:
  default-token-hg45s:
    Type:        Secret (a volume populated by a Secret)
    SecretName:  default-token-hg45s
    Optional:    false
QoS Class:       BestEffort
```

Condition **kubectl describe pods** *pod명*

Initialized	모든 초기화 컨테이너가 성공적으로 시작 완료
Ready	Pod는 요청을 실행 할 수 있음, 연결된 모든 서비스의 로드밸런싱 Pool에 추가되어야 한다는 뜻
ContainersReady	Pod 안 모든 컨테이너가 준비상태
PodScheduled	Pod가 하나의 노드로 스케쥴을 완료 했음
UnSchedulable	스케쥴러가 자원의 부족이나 다른 제약 등으로 지금 당장 Pod를 스케쥴 할 수 없음